

AI Usage DISCLOSURE FORM

1. Team Details

Team Name: Byte Pair

Project / Product Name: AI-Powered Code Retrieval System

Organization / Institution (if any): Ramiah Institute of Technology

Submission Date: 30 September 2026

2. AI Usage Declaration

Did your team use any Artificial Intelligence (AI) in developing this project?

Yes

3. Purpose of AI Usage (Brief Details)

Idea generation / brainstorming: Used AI to discuss and refine the approach for natural-language code retrieval, preprocessing, indexing, embedding-based search, and the overall system workflow.

Code generation or assistance: Used AI assistance for developing and modifying parts of the Python implementation, retrieval pipeline, indexing scripts, demo scripts, and supporting code.

UI / UX design: Not applicable

Content creation: Used AI assistance to prepare documentation, explanations, presentation content, and the demonstration video script.

Data analysis: Used AI assistance to understand retrieval results and analyze the behavior of the retrieval pipeline.

Testing / debugging : Used AI assistance to identify and resolve implementation issues, including Python environment/setup issues, file/index paths, and execution/debugging problems.

Other: AI was also used as a development assistant for understanding the problem statement, interpreting errors, improving implementation decisions, and explaining the final system.

4. Feature Origin Classification (Please add/delete based on the number of the feature)

1. Feature Name: Code Retrieval Pipeline

2. Self-Generated / AI-Generated / Both: Both

3. Description:

The team designed the overall code retrieval workflow based on the hackathon problem statement. AI tools were used as development assistance to discuss possible retrieval approaches, implementation details, and debugging. The final implementation was assembled, tested, and modified by the team. The system takes a natural-language query and retrieves relevant code snippets based on their semantic similarity.

1. Feature Name: Pre-indexed Code Retrieval

2. Self-Generated / AI-Generated / Both: Both

3. Description:

The team implemented a pre-indexing approach so that the code corpus is embedded and indexed ahead of query time. This avoids rebuilding the complete index for every query and allows the live demo to perform retrieval efficiently. AI assistance was used to help implement, troubleshoot, and explain the indexing and retrieval workflow. The team configured and tested the final index and integrated it with the query-time pipeline.

1. Feature Name: Embedding-Based Semantic Search

2. Self-Generated / AI-Generated / Both: Both

3. Description:

The system uses an embedding model to represent the natural-language query and code snippets in a shared semantic space and retrieves code based on relevance. AI tools were used to assist with implementation and debugging of the embedding and retrieval pipeline. The team selected, integrated, tested, and modified the model and retrieval implementation for the project requirements.

1. Feature Name: CPU-Based Query-Time Inference

2. Self-Generated / AI-Generated / Both: Both

3. Description:

The solution was designed to perform the live query and retrieval stage on CPU, in line with the requirement that the retrieval solution should be capable of running on CPU with minimal GPU resource utilization. AI assistance was used to separate the one-time index-building process from query-time inference and to debug the CPU execution workflow. The final CPU-based demo configuration was tested by the team.

1. Feature Name: Live Query Demonstration

2. Self-Generated / AI-Generated / Both: Both

3. Description:

The team created a live demonstration workflow where a natural-language programming query is provided and the system returns relevant code snippets. AI was used to assist with the demo script, command-line workflow, and debugging. The team modified and tested the implementation and prepared the final demonstration. This directly demonstrates the required retrieval behavior described in the challenge guidelines.

5. Ethical & Compliance Confirmation

AI usage complies with guidelines and policies. **Yes**

No proprietary or copyrighted data misused. **I Agree**

6. Declaration & Sign-Off

Name of Team Representative: Saikiran Shet

Role: Team member

Signature:



Date: 30/09/2026